Of course in many species the father does work hard and faithfully at looking after the young. But even so, we must expect that there will normally be some evolutionary pressure on males to invest a little bit less in each child, and to try to have more children by different wives. By this I simply mean that there will be a tendency for genes that say 'Body, if you are male leave your mate a little bit earlier than my rival allele would have you do, and look for another female', to be successful in the gene pool. The extent to which this evolutionary pressure actually prevails in practice varies greatly from species to species. In many, for example in the birds of paradise, the female receives no help at all from any male, and she rears her children on her own. Other species such as kittiwakes form monogamous pairbonds of exemplary fidelity, and both partners cooperate in the work of bringing up children. Here we must suppose that some evolutionary counter-pressure has been at work: there must be a penalty attached to the selfish mate-exploitation strategy as well as a benefit, and in kittiwakes the penalty outweighs the benefit. It will in any case only pay a father to desert his wife and child if the wife has a reasonable chance of rearing the child on her own.

Trivers has considered the possible courses of action open to a mother who has been deserted by her mate. Best of all for her would be to try to deceive another male into adopting her child, 'thinking' it is his own. This might not be too difficult if it is still a foetus, not yet born. Of course, while the child bears half her genes, it bears no genes at all from the gullible step-father. Natural selection would severely penalize such gullibility in males and indeed would favour males who took active steps to kill any potential step-children as soon as they mated with a new wife. This is very probably the explanation of the so-called Bruce effect: male mice secrete a chemical which when smelt by a pregnant female can cause her to abort. She only aborts if the smell is different from that of her former mate. In this way a male mouse destroys his potential step-children, and renders his new wife receptive to his own sexual advances. Ardrey, incidentally, sees the Bruce effect as a population control mechanism! A similar example is that of male lions, who, when newly arrived in a pride, sometimes murder existing cubs, presumably because these are not their own children.

A male can achieve the same result without necessarily killing step-children. He can enforce a period of prolonged courtship before he copulates with a female, driving away all other males who approach her,